Fingerprint-Based Voting System

A Secure Biometric Solution
for Elections

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Votes Can’t Lie… But Voters Sometimes Can

- Traditional EVMs allow impersonation & duplicates.
- Weak ID verification can causes fake voting.
- Also, paper systems can cause delays & human error.

Motivation

- It eliminates proxy & duplicate voting.
- It builds trust through verifiable authentication.
- It is designed for scalability & transparency.

"Secure. Transparent. One Finger, One Vote."

Objectives

Prevent Voter Impersonation

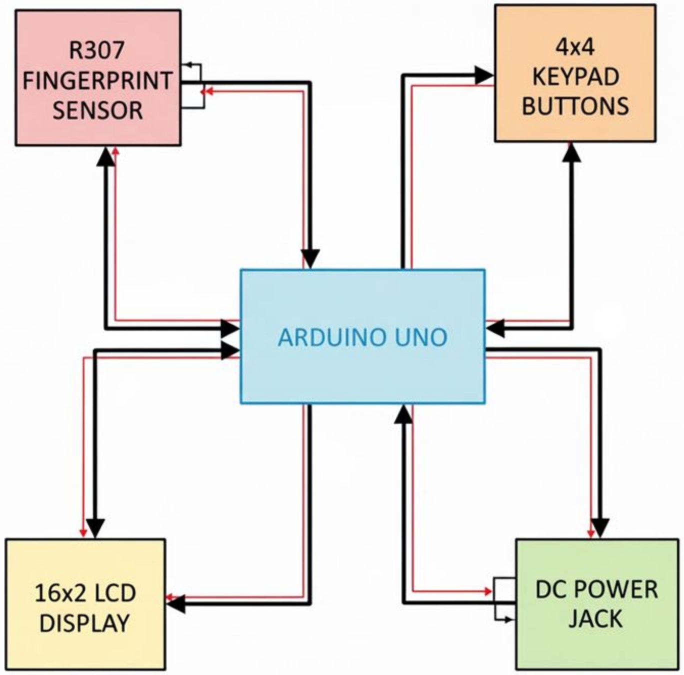
Improve Transparency and Trust

Simplify Voter Verification

Scalability for Future Use

Promote Cost-Effectiveness

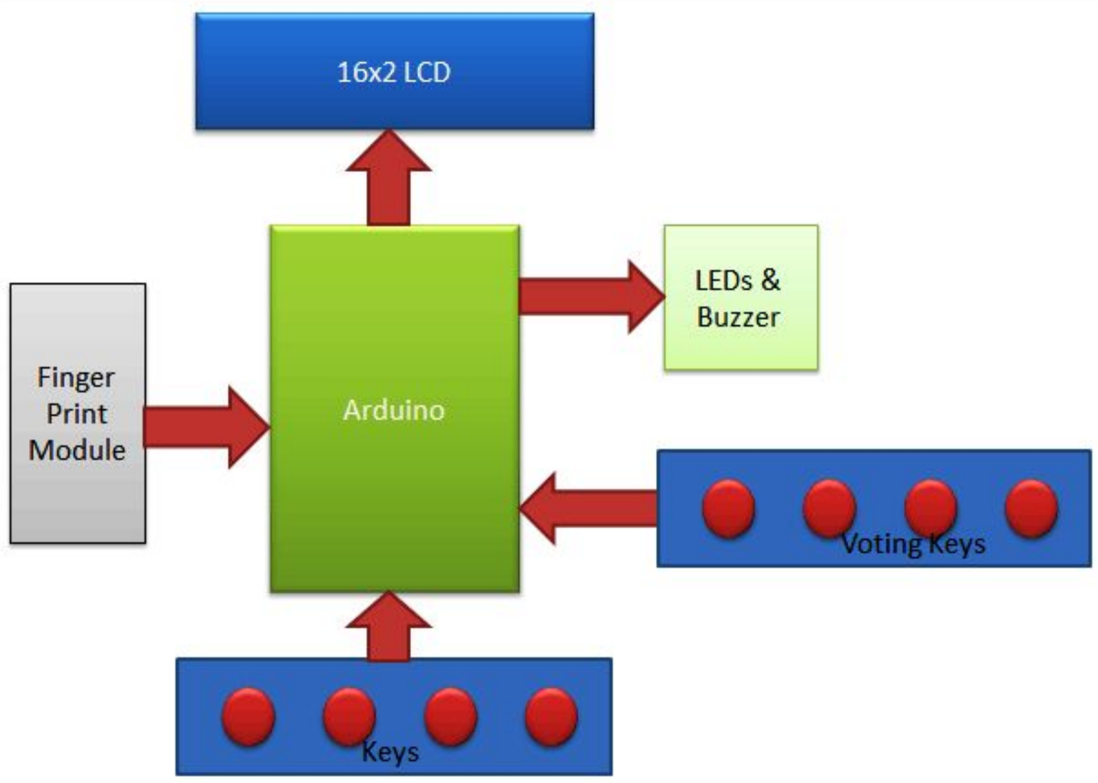
System Overview



- **Central Processing Through Arduino Uno**
  The Arduino Uno functions as the main controller, coordinating all input and output operations within the voting system.
- **Biometric Authentication via R307 Sensor**
  The fingerprint sensor captures and sends biometric data to the Arduino, ensuring that only authorized and enrolled voters can access the voting interface.
- **User Interaction Through Button Inputs**
  Push buttons serve as the primary user interface, enabling key operations such as enrollment, verification, deletion, and vote casting.

- **Real-Time Feedback Through LCD Display**
  A 16×2 I²C LCD module provides instant system messages, instructions, and voting results to guide the voter and admin during operation.
- **Stable Operation Powered Through DC Input**
  A regulated DC power supply feeds the Arduino, which further distributes the required voltage to all connected modules for reliable system performance.

Block Diagram



Hardware Components

Arduino uno

fingerprint sensor R-307

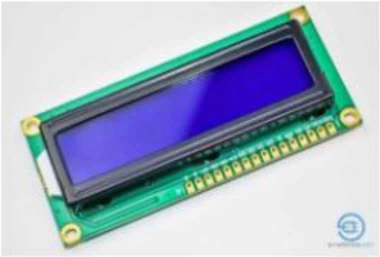
push button

jumper wires

LED

buzzer

LCD



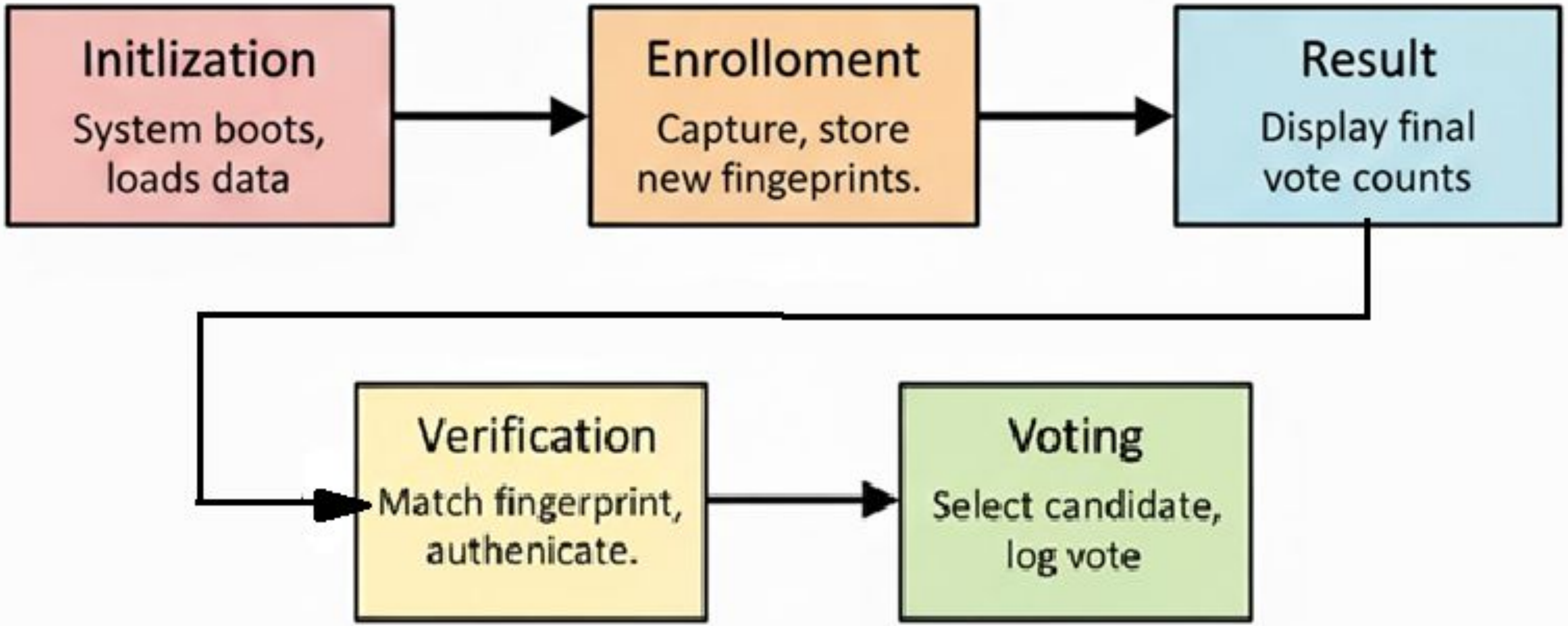
Breadboard

resistor

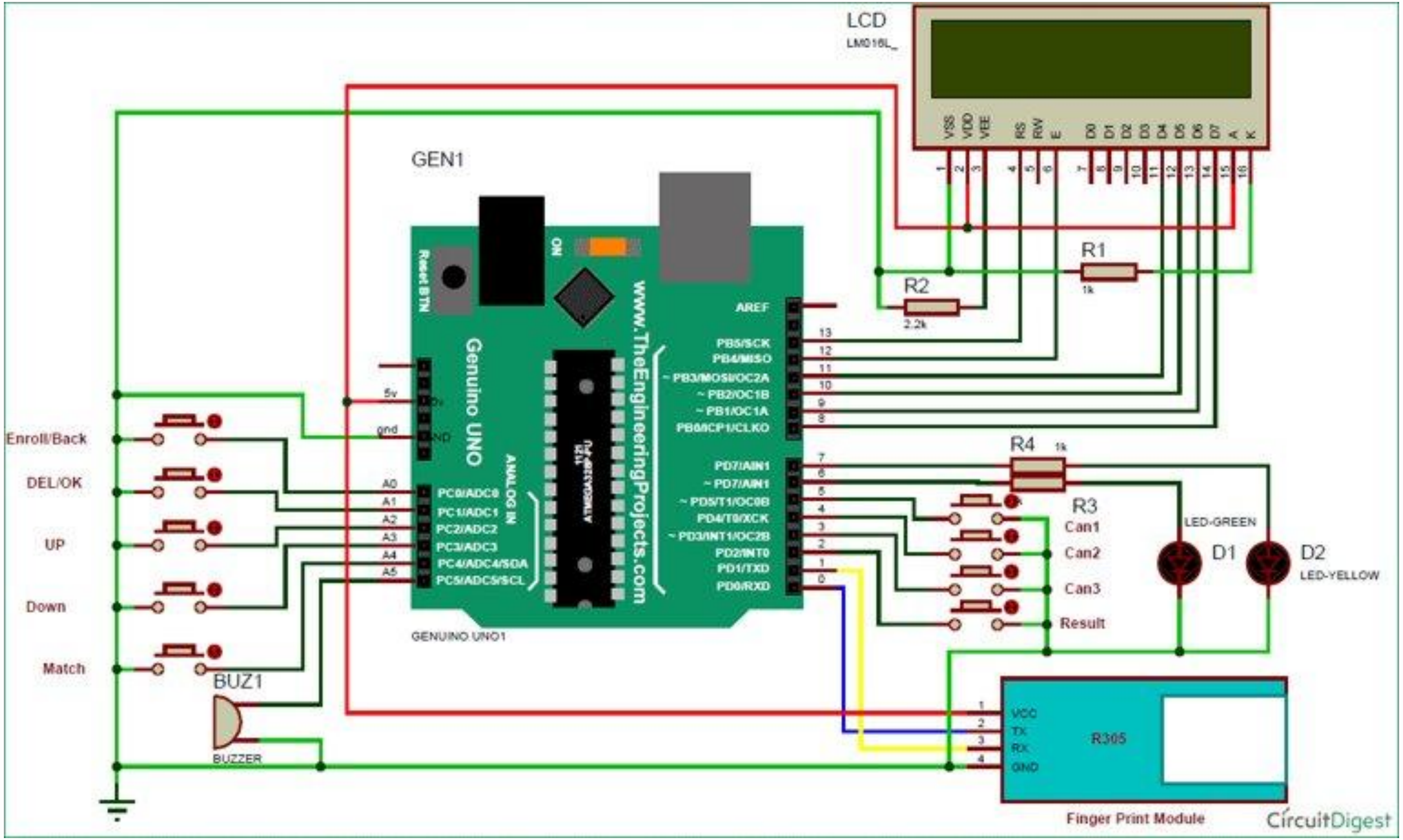
Software Architecture

Technologies & Methods Used:-

- **Programming Language:** C++ using the **Arduino IDE**
- **Biometric Processing:** Onboard **DSP (Digital Signal Processing)** of the R307 for image processing & template matching
- **Data Storage:** Internal **EEPROM** for vote counts and voting status
- **I/O Handling:** Interrupt-like button polling for user operations
- **Display Interface:** I^2C LCD for user feedback and result visualization


Initlization
System boots, loads data
Enrolloment
Capture, store new fingeprints.
Result
Display final vote counts
Verification
Match fingerprint, authenicate.
Voting
Select candidate, log vote

Circuit Description



Working Principle

1. **Initialization:** The Arduino Uno boots, initializes the fingerprint sensor, LCD display, and loads stored data from EEPROM.
2. **Enrollment:** New voter fingerprints are captured, processed using the sensor's internal DSP algorithm, and stored in both the R307 module and the EEPROM for tracking who has/has not voted.
3. **Verification:** Fingerprint images are matched against stored templates using the R307's onboard DSP matching engine. Only authenticated voters proceed.

4. **Voting:** Verified users select their desired candidate via push buttons. The Arduino logs each vote and updates the EEPROM securely.

5. **Result:** The system retrieves cumulative vote counts from EEPROM and displays the final results on the LCD when triggered by the admin.

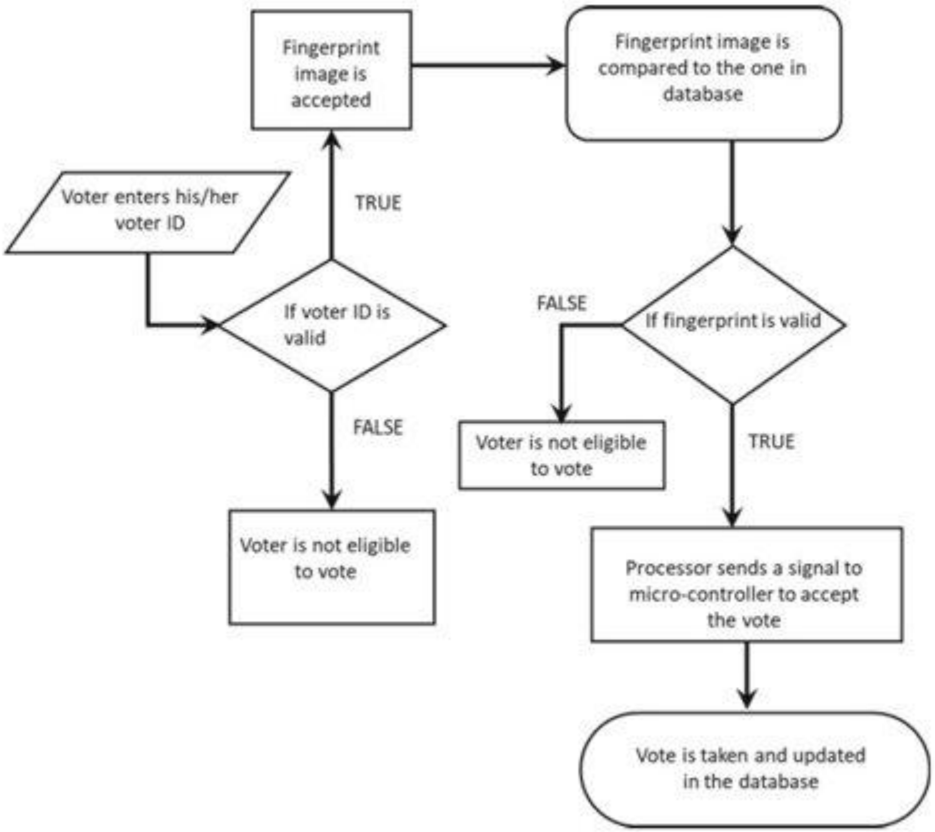
Key Features

- 127 fingerprints capacity
- 9 buttons (registration + Candidates)

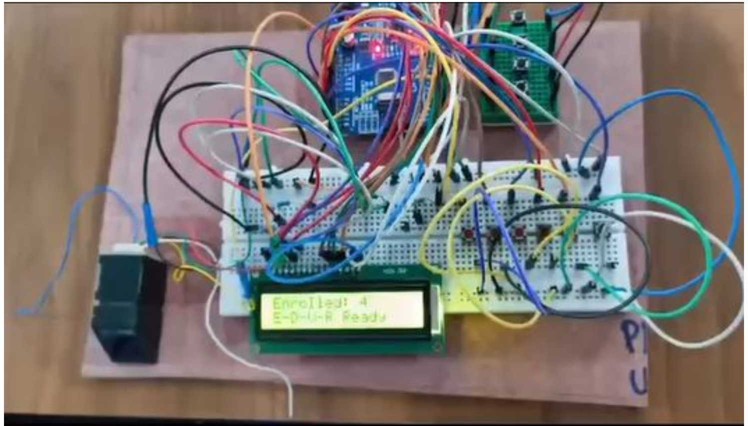
BUTTONS

3 for the candidates
5 for the registration
1 for initialization

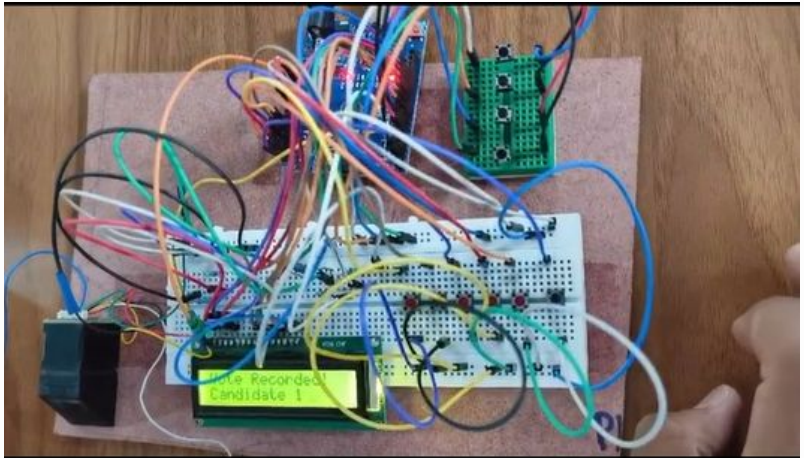
Algorithm / Workflow



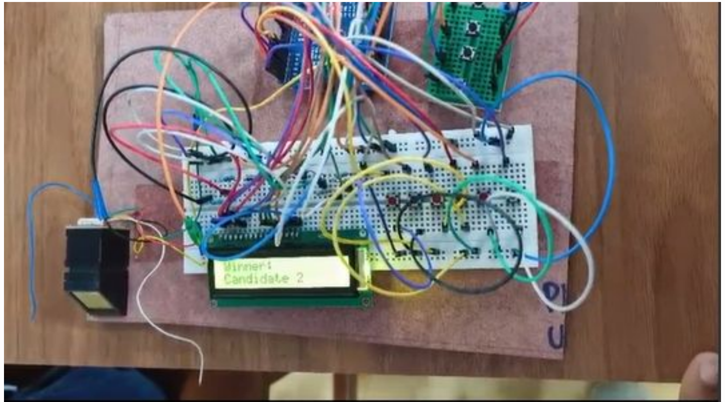
Result & Output



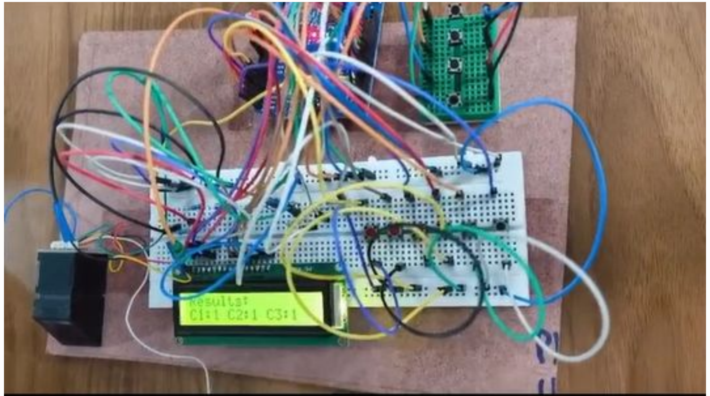
Enrollment process



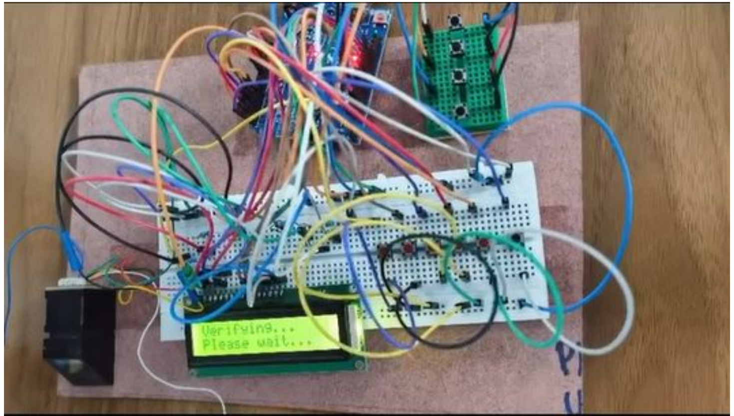
Voting



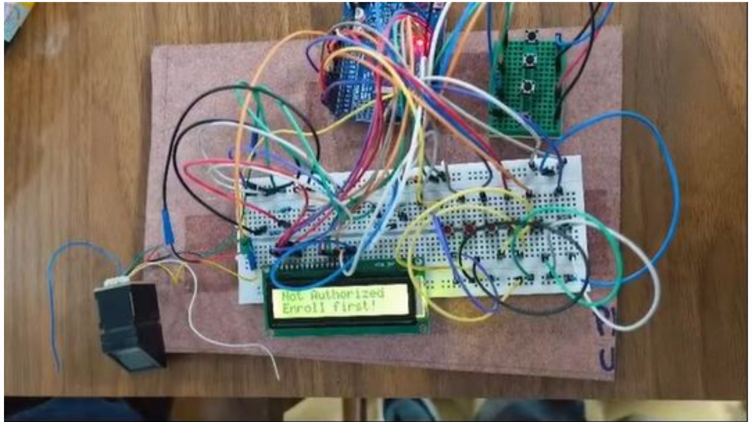
Results



Results (TIE)

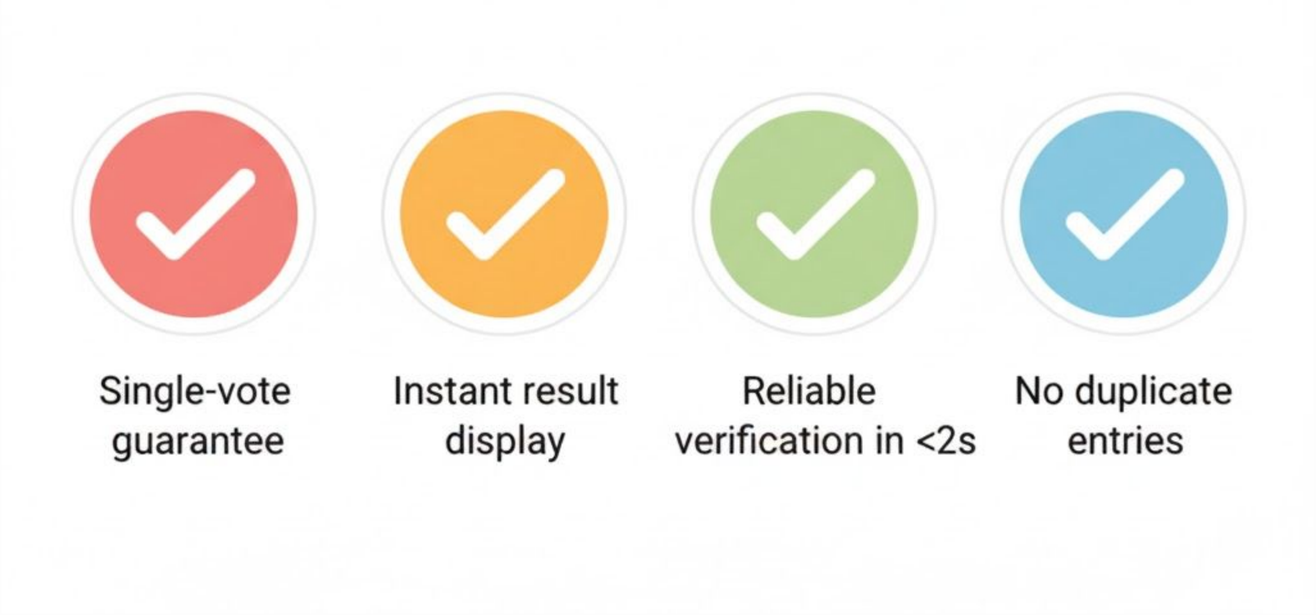


Verification (authorized)



Verification(not authorized)

Key Findings



Advantages/Disadvantages:-

Advantages:-	Disadvantages:-
High security: prevents impersonation and bogus voting	Hardware dependency: sensors/microcontrollers may malfunction
One person, one vote: fingerprints are unique	Biometric errors: elderly or manual laborers may have unreadable fingerprints
Fast & reliable authentication	Database management: storing fingerprint data securely is critical

Digital record keeping: faster counting, less human error.	Power & connectivity issues: needs reliable power, possibly internet
Tamper resistance: adds extra protection	Privacy concerns: voters may worry about misuse of biometric data
Transparency & trust: increases voter confidence	Initial setup costs can be high at large scale
Low-cost prototype: affordable for academic/demo use	Technical literacy: rural or less tech-savvy voters may need guidance

Future Enhancements

Conclusion

- The prototype demonstrates scalable, secure voting
- Enhances election transparency
- Ready for real-world adaptation and improvement

THANK YOU